

Adaptive Hybrid Framework for Multiscale Void Inspection of Chip Resistor Solder Joints

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Abstract—During reflow soldering, voids inevitably emerge inside the solder joints of chip resistors, which will influence the reliability of the electronic device. In this article, an adaptive hybrid framework is proposed for multiscale void inspection of chip resistor solder joints. First, an adaptive partition scheme is proposed to adaptively divide the image of the chip resistor solder joint into two regions, which correspond to the solder beneath the chip resistor body (Region 1) and the solder fillet (Region 2). Then, the voids in the two regions are separately inspected by two proposed adaptive inspection methods. One is that the multiscale convolution with adaptive circular kernels (MC-ACKs), guided filtering, and thresholding segmentation is proposed to inspect the voids in Region 1. The other is that an improved active contour model (ACM) is proposed to inspect the voids in Region 2, which incorporates the gray adaptive strategy and shape prior (SP) of the void. Finally, the shape factor and average gray strategy are utilized for void fine inspection of the chip resistor solder joint. Experimental results indicate that the proposed framework is superior to some existing methods, with an inspection performance of 0.9069 average Dice coefficient, 0.9907 F1 score, and 98.51% accuracy at a reasonable inspection speed.

Index Terms—Active contour model (ACM), average gray strategy, chip resistor, multiscale convolution with adaptive circular kernels (MC-ACKs), void detection.

I. INTRODUCTION

CHIP resistor has been widely used in surface mount technology (SMT) due to its small size, lightweight, high mechanical strength, and low assembly cost, which is beneficial for the reliability and stability of circuits [1]. Chip resistors can be mounted on the printed circuit board (PCB) via reflow soldering, which is one of the soldering methods [2]. Some voids will emerge in the solder joints of the chip resistor,

if the gas is trapped in the solders during reflow soldering [3]. Compared with those inside the ball grid array (BGA), the voids inside the chip resistor solder joint will result in more serious influences on the component reliability, such as the increase of its thermal resistance and the degrading of PCB mechanical strength [4]. Thus, void inspection is indispensable for chip resistors during the SMT process in real industries. Commonly, X-ray imaging is applied to acquire chip resistor images involving the voids inside their solder joints, due to its ability of nondestructive detection [5]. These images are observed by quality control (QC) workers to inspect voids, which is a low-efficient job. Also, long-term inspection will bring out visual fatigue to workers, causing false and missing inspection [6]. To deal with these problems inherently existing in manual inspection, the automatic X-rayed void inspection (AXVI) instrument is an alternative and promising means to high-efficiently ensure the reliability of mounted chips. For void inspection in the solder joints, the AXVI instrument can automatically acquire the images of solder joints, position the solder joints' regions, and inspect the voids inside solder joints [7]. High-performance void inspection algorithms are greatly important for the AXVI instrument, which restrict the performance of the instrument [8]. Due to the complex background, poor contrast, and porosity, void inspection is still an open and challenging problem for the AXVI instrument [9].

Some void inspection methods have been successfully applied to BGA components, in which traditional image processing [10], [11], [12], [13], [14], [15], [16], [17] and deep learning [18], [19] are involved. Sa-nguannam and Srinonchat [10] extracted solder balls from the BGA image via an adaptive binarization segmentation and considered that the voids demonstrated white regions in solder balls. Zhang et al. [11] extracted white regions as voids from the horizontal cross section images of 3-D X-rayed BGA solder joints, which employed a fixed threshold scheme. They also calculated the void content for each BGA component. Said et al. [12] used template matching to extract BGA solder balls, based on their shapes, areas, and regular distribution grids. Then, voids were inspected by subsequently Laplacian of Gaussian (LoG) edge detection and void distinguishing according to shape factor and average gray. Finally, the void content was calculated for each BGA component. Peng and Nam [13] proposed a new Blob analysis method to inspect and calculate the void content for the BGA component. Blob images were generated by the gradients of the filtered BGA solder joint images via a multiscale box filter. Void content was calculated after the voids were inspected based on mathematical

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morphology. Mouri et al. [14] defined void inspection as a matrix decomposition problem, in which voids were assumed as sparse components. Nonnegative matrix factorization was used to separate the void region from the smoothed BGA solder ball image via three-image average filtering. Nuanprasert et al. [15] combined Hough transform and Delaunay triangulation to extract BGA solder balls. Then, the LoG operator was applied to inspect voids in each solder ball after image contrast enhancement via histogram equalization. Le et al. [16] proposed a void detection method according to the characteristics of solder joint voids, after obstructed solder joints were detected by an interactive X-ray contour extraction method. Their method is a combination of a series of traditional image processing methods, such as grayscale morphology, histogram stretching, fuzzy enhancement, and Blob analysis. Ahuja and Neeluru [17] detected BGA solder balls via the local adaptive threshold method and template matching. Then, LoG combined with radial and angular direction scanning was employed to inspect the voids in solder balls. Because of the rapid development of deep learning in recent years, some researchers applied it to inspect the voids inside BGA solder balls. Akdeniz et al. [18] designed a full convolution neural network (FCNN) for feature extraction and cascaded a minimum distance classifier to internal defect classification for BGA solder balls. Essentially, they classified BGA solder balls with defects into four categories rather than inspecting the voids, which was impossible to calculate the void content. Neeluru and Ahuja [19] employed a U-Net to inspect the voids and subsequently calculated the void content of each BGA solder ball, in which the train data were augmented based on the priori knowledge of voids.

Although the above methods have performed the inspection of voids in BGA components, some difficulties and challenges should be dealt with if they are applied to inspect the voids inside chip resistor solder joints. First, the images of chip resistor solder joints demonstrate much more complexity than the images of BGA solder balls, in which the voids in different regions of the mounted chip resistor demonstrate quite different characteristics, such as the density, contrasts, and sizes of the voids, which will be discussed in detail in Section II. Thus, the global operations used in traditional image processing, such as LoG and global thresholding, are not employed to process the images of chip resistor solder joints, which indicates that these traditional image processing is not suitable for inspecting the voids inside chip resistor solder joints. Second, due to the data-driven characteristics of deep learning, deep learning-based inspection methods require a large amount of manually annotated data for learning intrinsic patterns of the defects for subsequent inspection. However, the annotation is significantly time-consuming since the voids are quite different in size and contrast, and so many voids exist in the images. Thus, this arises a challenging problem for the flexible manufacturing with the characteristics of multispecification, small batch, and high turnover, which is becoming a trend in manufacturing.

In this article, an adaptive hybrid framework is proposed for multiscale void inspection of chip resistor solder joints. To the best of our knowledge, this is the first attempt of void inspection for chip resistor solder joints. First, an adaptive partition

scheme is proposed to adaptively divide the chip resistor solder joint into two regions, corresponding to the solder beneath the chip resistor body and the solder fillet. Then, considering the different characteristics of the voids in the two regions, two adaptive inspection schemes are proposed to inspect the voids separately and coarsely in the two regions. One is to design a novel active contour model (ACM) incorporating the improved gray adaptive (IGA) strategy and shape prior (SP) of the void, termed as an abbreviation of SP-IGA-ACM. The other is a combination of the proposed multiscale convolution with adaptive circular kernels (MC-ACKs), guided filtering, and thresholding segmentation. Finally, the voids are finely inspected by the shape factor and the average gray strategy.

To summarize, the main contributions of this work are described as follows.

- 1) We propose an adaptive hybrid framework with a multiscale strategy and several adaptive schemes to perform a coarse-to-fine void inspection task for chip resistor solder joints, which is beneficial for the AXVI instrument.
- 2) An optimization problem is formulated to adaptively divide the image of the chip resistor solder joint into the regions of the solder beneath the chip resistor body and the solder fillet for subsequent void coarse inspection.
- 3) The MC-ACK scheme is proposed to coarsely inspect the voids inside the solder beneath the chip resistor body, which utilizes the MCs with different circular kernels adaptively determined for each chip resistor solder joint.
- 4) The SP-IGA-ACM scheme is proposed to coarsely inspect the voids inside the solder fillet, which combines the proposed gray adaptive strategy to improve the time efficiency of the ACM and an SP strategy to accurately inspect overlapping voids.

II. IMAGE ACQUISITION AND IMAGE DATASET

X-ray imaging is widely applied to the observation of internal structures of electronic components, which involves 2-D X-ray imaging and 3-D X-ray imaging. Although 3-D X-ray imaging has a micrometer resolution and precisely measures the object in 3-D, it spends much time reconstructing the 3-D object [20], which is not acceptable for real electronic industries. Comparatively, 2-D X-ray can quickly acquire the internal information of electronic components, which is usually applied to inspect the internal defects of electronic components in SMT industries [21]. In this article, 134 X-rayed images of chip resistor solder joints are acquired by a 2-D X-ray perspective analyzer, whose sizes range from 190×210 pixel to 350×400 pixel.

Fig. 1 illustrates a mounted chip resistor image acquired by the 2-D X-ray perspective analyzer, which involves two solder joints. Since the voids emerge inside chip resistor solder joints, a solder joint can be automatically separated from the chip resistor image. When the chip resistor is mounted on the substrate by the reflow soldering processing, some solder is filled beneath the chip resistor body and other solder forms the solder fillet. The thickness of the solder beneath the chip resistor body is different from that of the solder fillet and the lead in the solder has a high attenuation to X-ray [22]. Thus, this thickness difference results in the X-ray perspective

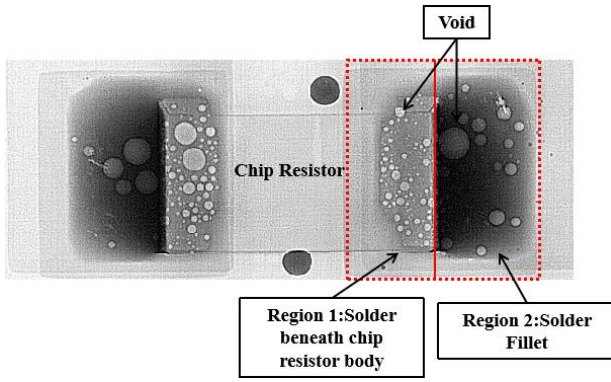


Fig. 1. Chip resistor X-ray image.

difference and further results that the image chip resistor solder joint demonstrates two regions with saliently different image characteristics, as labeled by red rectangles in Fig. 1. For simple descriptions, we labeled Regions 1 and 2 to correspond to the solder beneath the chip resistor body and the solder fillet, respectively. The background is relatively dark and the voids are obscure in Region 2, while the background is relatively bright and the voids are salient in Region 1. Furthermore, since the area of the solder determines the length of the escape path of the gas in the reflow soldering process [23], the voids in Region 2 demonstrate relatively large shapes and sparse distribution, while those are relatively small and densely distributed in Region 1. Also, the voids demonstrate quite different sizes and different contrasts, which indicates that global and single-scale operations possibly meet with the difficulty for accurate void inspection.

III. PROPOSED METHOD

A. Architecture of the Inspection Framework

As discussed in Section II, the image of the chip resistor solder joint can be visually divided into two regions due to the X-ray imaging characteristics and the reflow soldering process. The two regions demonstrate quite different image characteristics in terms of contrast, void size, and void distribution. The facts inspire us to partition the image of the chip resistor solder joint into two regions similarly illustrated in Fig. 1. Thus, we can elaborately design different inspection schemes to well inspect different regions according to their image characteristics. In this article, we have proposed an adaptive hybrid framework to implement this assumption for void inspection of chip resistor solder joints, whose architecture is sketched in Fig. 2.

The proposed framework can be divided into three stages, which are adaptive partition, void coarse inspection including an improved ACM scheme and an MC scheme, and void fine inspection. Due to salient differences of image characteristics between the two regions of the chip resistor solder joint, an adaptive partition scheme is proposed to adaptively divide the chip resistor solder joint image into two regions, labeled as Regions 1 and 2 indicated in Fig. 1, each of which can be processed by an appropriate inspection scheme to improve the inspection ability of the framework. Since the voids in

Region 2 have the characteristics of large size, sparse distribution, and low contrast, we introduce the ACM to inspect these voids due to its excellent subpixel segmentation ability [24]. Here, an improved ACM scheme termed as SP-IGA-ACM is proposed to improve the time efficiency of the ACM. Different from the voids in Region 2, the voids in Region 1 have the characteristics of small size and dense distribution. If the ACM-based scheme is employed for inspection, the evolving curve is possibly ignored during evolution due to the small sizes of the voids. This means that small voids will most likely not be inspected. Thus, an MC scheme termed MC-ACK is proposed to inspect the voids in Region 1, which is followed by combined with guided filtering and Otsu. Finally, the inspected chip resistor solder joint image at the stage of coarse inspection is postprocessed by shape factor and the average-gray strategy for fine inspection, which can effectively eliminate void misinspection and improve the robustness of the inspection framework.

B. Adaptive Partition

Due to the different image characteristics of Regions 1 and 2, two different schemes should be designed to inspect the voids in different regions, which can improve the inspection ability of the proposed framework. Since the chip resistor demonstrates a rectangle appearance, the boundary between Regions 1 and 2 is theoretically a straight line. Thus, an adaptive partition scheme is proposed to find this straight line $\hat{i}$ for partitioning the chip resistor solder joint image into two regions, which is formulated as

$$\begin{aligned} \hat{i} &= \arg \max \left| \frac{I_i^l(x)}{\text{row} * \text{col}^l(i)} - \frac{I_i^r(x)}{\text{row} * \text{col}^r(i)} \right| \\ \text{s.t. } \text{col}^l(i) + \text{col}^r(i) &= \frac{N}{2} \\ \frac{N}{4} &\leq i \leq \frac{3N}{4} \end{aligned} \quad (1)$$

where $I_i^l(x)$ and $I_i^r(x)$ represent the total gray values of the left and right regions at the division line i , respectively. $\text{col}^l(i)$ and $\text{col}^r(i)$ mean the numbers of columns of the corresponding regions, respectively. N and row are the total columns and rows of the chip resistor solder joint image, respectively.

The optimization problem (1) is solved by uniform ergodic to obtain the approximate optimal solution $\hat{i}$. Thus, the image can be adaptively divided into the two regions via $\hat{i}$, which correspond to solder beneath chip resistor body and solder fillet, denoted as I_{R1} and I_{R2} , respectively.

C. Void Coarse Inspection

After adaptive partition, the chip resistor solder joint image can be adaptively divided into two regions. We propose two adaptive schemes to coarsely inspect the voids in the two regions according to their different image characteristics. That is, a novel multiscale circular convolution scheme termed MC-ACK and an improved ACM scheme termed SP-IGA-ACM are proposed to coarsely inspect the voids in Regions 1 and 2, respectively.

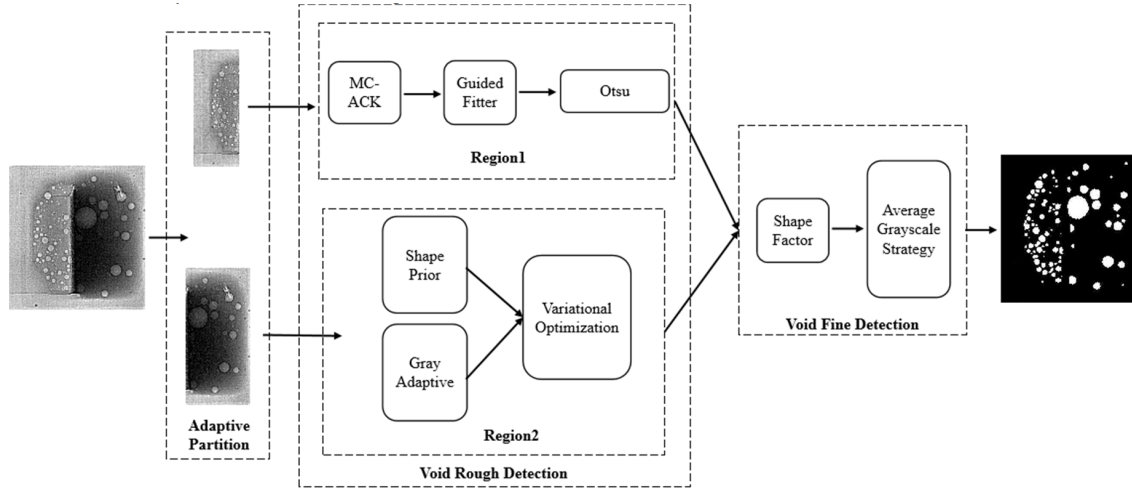


Fig. 2. Architecture of the proposed inspection framework.

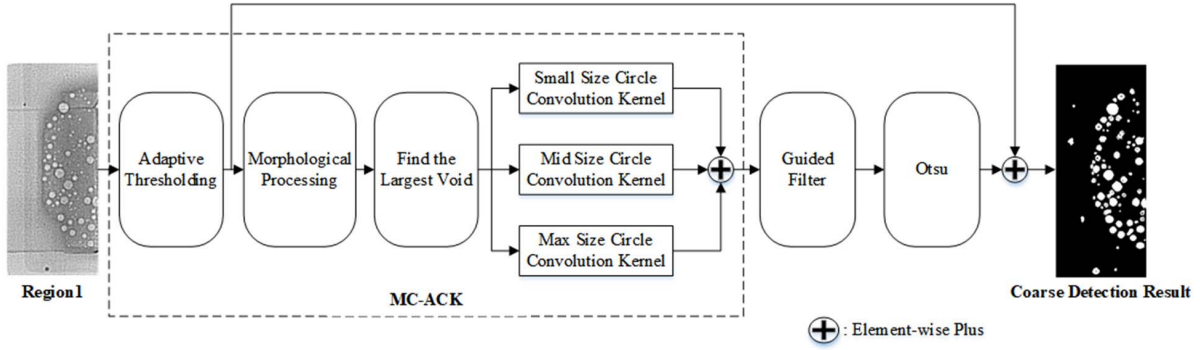


Fig. 3. Flowchart of coarse inspection for Region 1.

1) *Multiscale Convolution With Adaptive Circular Kernels*: Since the voids demonstrate circle-like appearances and Region 1 has high image contrast, we propose a novel multiscale circular convolution scheme termed MC-ACK to adaptively deal with the voids with different sizes. As illustrated in Fig. 3, the MC-ACK is combined with guided filtering [25] and thresholding via Otsu to inspect the voids in Region 1. The proposed MC-ACK is described in detail as follows.

After Region 1 of the chip resistor solder joint image is processed by adaptive thresholding [26] and mathematical morphology, the image can be processed by the circular convolution with a fixed kernel. However, the voids inside different solder joints have quite different appearances illustrated in the images, which indicates that the fixed circular kernel cannot be suitable in practice. Here, we first propose an ACK strategy for circular convolution, in which the size of the circular kernel is adaptively determined by the largest void in Region 1 of each chip resistor solder joint, which is defined as

$$W_j^{la} = 2\text{int}\left(\sqrt{\frac{S_j^{la}}{\pi}}\right) + 1, j = 1, 2, \dots, \text{Total} \quad (2)$$

where S_j^{la} is the area of the largest void in the j th chip resistor solder joint image and Total is the number of the

chip resistor solder joint images. W_j^{la} denotes the large-scale circular convolution for the corresponding image. As indicated in (2), different chip resistor solder joints have the large-scale convolutions with different circular kernels. However, when the voids inside a chip resistor solder joint are greatly different in size, small voids are probably not inspected by the ACK strategy. Thus, we further propose the MC-ACK strategy to deal with this problem, including large-scale, small-scale, and midscale circular convolutions. Since the minimal void can be predetermined according to the QC standard in real industries, the size of the corresponding circular kernel W_j^{sm} for the small-scale convolution can be defined as similar to (2), and the size of the circular kernel W_j^{mi} for the midscale convolution can be defined as

$$W_j^{mi} = \text{int}\left(\frac{W_j^{sm} + W_j^{la}}{2}\right). \quad (3)$$

In this article, $W_j^{sm} = 3$. It is noted that the kernel sizes for the large-scale and midscale convolutions are adaptively determined and different for each chip resistor solder joint, while that for the small-scale convolution is fixed for all the chip resistor solder joint when the QC standard is ensured in real industries. This simple adaptive strategy is acceptable and highly efficient for real industries, since the voids in Region 1

have relatively small differences in size compared with those in Region 2.

Thus, the (a, b) th element $\mathbf{O}_{W_j^{\text{ms}}}(a, b)$ in the circular kernel for the MC can be designed as

$$\mathbf{O}_{W_j^{\text{ms}}}(a, b) = -\varepsilon \left(\left(\frac{W_j^{\text{ms}}}{2} \right)^2 - (|a - A|^2 + |b - B|^2) \right) \quad (4)$$

$a \neq A, b \neq B, \text{ms} = \text{la, mi, sm}$

where (A, B) represents the location of the center element of the kernel, and the corresponding value is the absolute value of the sum of all the elements of the kernel except for the center element. It indicates that the sum of the values of all the kernel elements equals zero. $\varepsilon(t)$ represents a step function.

Here, we illustrate an example of $W_j^{\text{ms}} = 5$ for the design of the corresponding circular kernel $\mathbf{O}_5$ as

$$\mathbf{O}_5 = \begin{bmatrix} 0 & -1 & -1 & -1 & 0 \\ -1 & -1 & -1 & -1 & -1 \\ -1 & -1 & 20 & -1 & -1 \\ -1 & -1 & -1 & -1 & -1 \\ 0 & -1 & -1 & -1 & 0 \end{bmatrix}. \quad (5)$$

It can be seen that the shape of the kernel $\mathbf{O}_5$ is circle-like.

2) *Improved ACM With Grayscale Adaption and SP*: Due to the characteristics of low contrast, the above MC-ACK scheme cannot perform a good inspection of the voids in Region 2. Due to the advantages of multiscale and multitarget subpixel segmentation, high robustness, and high calculation efficiency, the local prefitting (LPF) ACM [27] is introduced into our proposed framework to coarsely inspect the voids in Region 2. However, the original LPF-ACM is poorly sensitive to the shape of the target, which will possibly limit its curve evolution to achieve the accurate contours of the overlapping voids. Also, it calculates LPF energy terms via a fixed sliding window, which will result in low time efficiency since some redundant computations emerge. To deal with the two problems, we modify the LPF ACM to propose an improved ACM incorporating an SP constraint term and an IGA strategy (SP-IGA-ACM), whose energy function is defined as

$$\begin{aligned} E^{\text{SP-GAACM}}(\varphi, e_1, e_2) = & \iint_{\Omega} K_{\sigma}(x - y) |I_{R2}^j(y) - e_1(x)|^2 \\ & \times H(\varphi(y)) dy dx \\ & + \iint_{\Omega} K_{\sigma}(x - y) |I_{R2}^j(y) - e_2(x)|^2 \\ & (1 - H(\varphi(y))) dy dx \\ & + v \int_{\Omega} \delta(\varphi) |\nabla \varphi(x)| dx \\ & + \mu \int_{\Omega} \frac{1}{2} (\nabla \varphi(x) - 1)^2 dx \\ & + \lambda \int_{\Omega} (\varphi - \psi)^2 dx \end{aligned} \quad (6)$$

where $\Omega \subset R^2$ represents the image domain, and $x, y \in \Omega$. I_{R2}^j denotes Region 2 of the j th chip resistor solder joint. K_{σ} is a Gaussian kernel function with standard deviation σ . In (6), the first data-driven term guides the level set function to evolve to the target edge. It is noted that the LPF functions $e_1(x)$ and $e_2(x)$ are adaptively calculated based

on the grayscale as illustrated in (8), which can improve the efficiency and accuracy of the level set function evolution. The second term $\int_{\Omega} \delta(\varphi) |\nabla \varphi(x)| dx$ is the length constraint term, which is employed to regularize the zero-level contour of φ . The third term $\int_{\Omega} (1/2) (\nabla \varphi(x) - 1)^2 dx$ is the distance regularizing term, which is employed to penalize the deviation of the level set function φ from a signed distance function in order to ensure the stable evolution of φ . The fourth term $\int_{\Omega} (\varphi - \psi)^2 dx$ is the SP constraint term, which can improve the sensitivity of the model to voids to improve the inspection performance for overlapping voids as illustrated in (10)–(12). v is a weight coefficient of the length constraint term to control the length of the contour curve. Specifically, a large v can shorten the length of φ but cause inadequate segmentation; and a small v may lead to the segmentation of undesirable background regions [28]. μ is a weight coefficient of the distance regularization term, which is related to the time step Δt of the evolution. The relationship is $\mu \times \Delta t \leq 0.1$ to satisfy the Courant Friedrichs Lewy (CFL) condition for numerical stability [29]. As a weight coefficient of the SP constraint term and given the balance among parameters, λ should be equal to the weight coefficient of the data-driven term. The Gaussian kernel function K_{σ} with standard deviation σ is truncated using a $(2k + 1) \times (2k + 1)$ window with $k \geq \sigma$, where a small σ makes the computation efficient with slow convergence, while a large σ increases computational burden and convergence rate [30]. These parameters have been sufficiently analyzed in previous studies [28], [29], [30], [31], which are also directly utilized for many applications. Thus, in this article, v, μ , and λ are assigned to $0.02 \times 255 \times 255, 2$, and 1 , correspondingly. Also, the parameters of Gaussian kernel function K_{σ} are set as $k = 4$ and $\sigma = 3$.

Meanwhile, the model formulated in (6) is not sensitive to the initial contour, which can be set arbitrarily. $H_{\varepsilon}(x)$ and $\delta_{\varepsilon}(x)$ are regularized Heaviside and Dirac functions, respectively, which is defined as

$$\begin{cases} H_{\varepsilon}(x) = \frac{1}{2} \left(1 + \frac{2}{\pi} \arctan\left(\frac{x}{\varepsilon}\right) \right) \\ \delta_{\varepsilon}(x) = \frac{1}{\pi(\varepsilon^2 + x^2)} \end{cases} \quad (7)$$

The LPF functions $e_1(x)$ and $e_2(x)$ are defined as

$$\begin{cases} e_m(x) = \text{mean}\left(I_{R2}^j(y)\right) y \in M \\ e_1(x) = \text{mean}\left(I_{R2}^j(y)\right) y \in M_1 \\ e_2(x) = \text{mean}\left(I_{R2}^j(y)\right) y \in M_2 \end{cases} \quad (8)$$

where $\begin{cases} M_1 = \{y | I_{R2}^j(y) < e_m(x)\} \cap M \\ M_2 = \{y | I_{R2}^j(y) > e_m(x)\} \cap M \end{cases}$

where M represents a sliding window. $e_m(x)$ is the average gray value of all the pixels in M . M_1 and M_2 are the regions, where the image gray values are smaller and larger than $e_m(x)$ of M , respectively. $e_1(x)$ and $e_2(x)$ represent the average gray values of all the pixels in M_1 and M_2 , respectively, which are calculated before the evolution of φ .

M influences $e_1(x)$ and $e_2(x)$ and further influences the evolution of φ . The fact indicates that the size of M influences

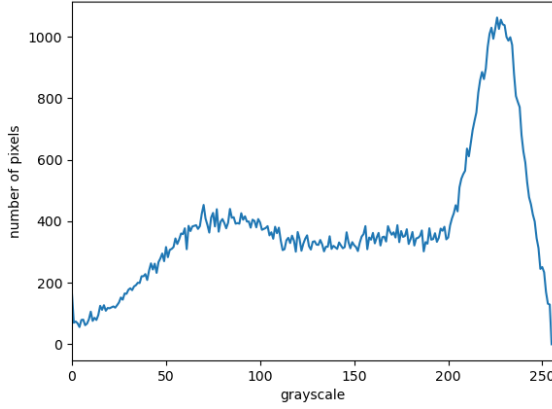


Fig. 4. Gray histogram of Region 2.

the segmentation performance in terms of time efficiency and segmentation accuracy. In the previous LPF ACM [27], M is fixed. If M is fixed as a small value, the segmentation accuracy for the region with low gray levels is worse, although the small window costs less time. If M is fixed as a large value, the large window costs more time and excessive segmentation easily occur in the region with high gray levels. To adaptively deal with these regions with different gray levels, we propose an IGA strategy to adaptively determine the window size M in (8), which is described as follows.

As shown in Fig. 4, the gray distribution of the pixels in Region 2 is relatively uniform in the range of [100, 200]. For simplicity, the gray values of the pixels in Region 2 can be roughly divided into three ranges, which are (0, 100), [100, 200], and (200, 255], respectively. Then, M can be adaptively determined as

$$M = T - 2 \left\lceil \frac{I_{R2}^j(x, y)}{100} \right\rceil \quad (9)$$

where $I_{R2}^j(x, y)$ denotes the gray value of the pixel in Region 2. $\lceil \cdot \rceil$ is a function of rounding-up to an integer. Thus, $1 < 2 \lceil (I_{R2}^j(x, y)/100) \rceil \leq 8$. Since the window size is odd, $M > 3$ and $M < 9$ are experimentally verified in this article. Combined with the ranges of $2 \lceil (I_{R2}^j(x, y)/100) \rceil$ and M , $T = 11$ is available.

The SP constraint term $\int_{\Omega} (\varphi - \psi)^2 dx$ in (6) is introduced to improve the inspection performance for overlapping voids, which is designed as follows. Since the void demonstrates a circle-like shape, a circular equation is utilized to simulate a void shape as

$$CE_i(x, y) = \frac{(x - x_i)^2 + (y - y_i)^2}{r_i^2} - 1 \quad (10)$$

where (x_i, y_i) and r_i are the center location and radius of the i th circle, respectively. These parameters are determined by applying morphological operations to a binary map of φ . Then, a binary SP can be defined according to (10) as

$$B_{\Psi}(x, y) = \begin{cases} 1, & CE_i(x, y) \leq 0 \\ 0, & CE_i(x, y) > 0. \end{cases} \quad (11)$$

Next, the SP is initialized as

$$\psi(x, y) = \begin{cases} 1, & B_{\Psi}(x, y) > 0 \\ -1, & B_{\Psi}(x, y) = 0. \end{cases} \quad (12)$$

During curve evolution, the SP is dynamically updated.

After the steepest descent method is utilized to minimize the energy function (6) [27], the following gradient descent flow can be achieved as:

$$\begin{aligned} \frac{\partial \varphi}{\partial t} = & -\delta_{\varepsilon}(\varphi)(e_s(x) - e_l(x)) + v\delta_{\varepsilon}(\varphi)\text{div}\left(\frac{\nabla \varphi}{|\nabla \varphi|}\right) \\ & + \mu\left(\nabla^2 \varphi - \text{div}\left(\frac{\nabla \varphi}{|\nabla \varphi|}\right)\right) + 2\lambda(\varphi - \psi) \end{aligned} \quad (13)$$

where $e_s(x)$ and $e_l(x)$ are

$$\begin{cases} e_s(x) = \int_{\Omega} K_{\sigma}(x - y)|I_{R2}^j(y) - e_1(x)|^2 dx \\ e_l(x) = \int_{\Omega} K_{\sigma}(x - y)|I_{R2}^j(y) - e_2(x)|^2 dx. \end{cases} \quad (14)$$

In the procedure of calculating (13), due to the distance regularization term in (6), spatial partial derivatives $(\partial \varphi / \partial x)$ and $(\partial \varphi / \partial y)$ are discretized as central differences, in which the space step is set to 1 [29]. It is noted that the space step influences the evolution accuracy and efficiency of φ . If it is too large, important attribute information of the evolved contour will be generalized; if too small, the evolution efficiency will be reduced. The temporal partial derivative $(\partial \varphi / \partial t)$ is discretized as a forward difference. Thus, (13) can be written as

$$\varphi^{n+1} = \varphi^n + \Delta t \cdot \frac{\partial \varphi^n}{\partial t} \quad (15)$$

where Δt is a time step. If Δt is too large, it will lead to a fast but jumpy evolution of φ and may cause a less detailed evolution for small objects; if too small, it will slow down the convergence rate. Meanwhile, Δt and μ are related by $\mu \times \Delta t < 0.1$ to satisfy the CFL condition for numerical stability [29]. In this study, Δt can be empirically set to 0.02. $(\partial \varphi^n / \partial t)$ is the approximation of the right-hand side in (13). The level set function φ is updated according to (13)–(15). To avoid the possible premature convergence caused by the small change of adjacent evolutions, the stopping criterion for (15) is set as $|(S_i - S_{i-2}/S)| < 10^{-5}$ [27], in which S , S_{i-2} , and S_i represent the area of Region 2, and the areas surrounded by the evolved contours at the $(i - 2)$ th and i th iterations, correspondingly. The initial contour is set as [0:80, 0:80]

$$P_{\text{seg}}^j = [P_{R1}^j : P_{R2}^j]. \quad (16)$$

The noise, such as background trace and vias [12], possibly emerges in P_{seg}^j , which will cause misinspection. Thus, void fine inspection should be implemented after void coarse inspection, as interpreted in Section III-D.

D. Void Fine Inspection

After coarse inspection, some misinspections, such as background trace and vias, emerge in the chip resistor solder joint image, which will influence the inspection performance and the calculation of void content. The appearance of the

background trace is not circle-like, which is quite different from those of voids and vias. So, the shape factor can be employed to suppress misinspection caused by background trace. Compared to the background, the vias demonstrate relatively dark, while the voids demonstrate relatively bright. The facts indicate that the voids and the vias can be distinguished by their average gray levels. So, fine inspection is performed on the coarsely inspected chip resistor solder joint image, which involves the strategies of shape factor and average gray, which is described as follows.

Assuming the area and circumference of the l th connected domain in P_{seg}^j as A_l and L_l , respectively, the corresponding shape factor SF_l is defined as

$$SF_l = \frac{4\pi A_l}{L_l^2}. \quad (17)$$

If the connected domain is close to the circle, SF_l will approach 1. Otherwise, it will approach 0. Misinspection can be suppressed by a threshold T_1 to obtain further inspection $P_{seg_2}^j$. In this article, if the area of the connected domain is larger than 60 pixels, T_1 is empirically set to 0.6. Otherwise, T_1 is empirically set to 0.8.

Although the background trace is suppressed, there are possibly some vias in $P_{seg_2}^j$. Since vias is circle-like, an average gray strategy is proposed to further suppress misinspection, which is defined as

$$\text{mean}(Rs_l) - \text{mean}(\text{Rec}_l - Rs_l) > T_2 \quad (18)$$

where Rs_l is the subimage of the chip resistor solder joint corresponding the l th connected domain in $P_{seg_2}^j$. Rec_l is the minimal bounding rectangle subimage of Rs_l . $\text{mean}(\cdot)$ is the average operation of the gray values of the pixels in the subimages. T_2 is a threshold, which is empirically set to 0.01. If (18) is satisfied, the l th-connected domain in $P_{seg_2}^j$ is considered as a void.

IV. EXPERIMENTAL RESULTS AND DISCUSSION

All the experiments are implemented by a computer with Windows 10 64-bit, six-core Intel i5-10600KF CPU at 4.10GHz. Simulations are programmed by Python3.

Three commonly used metrics are employed to evaluate the performance of the inspection methods for the voids inside chip resistor solder joints, which are accuracy rate (Acc), F1 score, and average Dice coefficient.

A. Influence of MC-ACK

The appropriate size of the circular convolution is significant for the coarse inspection of the voids in Region 1. Here, we conducted an experiment to validate the proposed ACK and MC-ACK by comparing them to the circular convolutions with different fixed kernels. It is noted that the proposed SP-IGA-ACM scheme is not involved and only Region 1 of each solder joint image is utilized for inspection in this experiment.

As indicated in Table I, with the increase of the kernel size, the average Dice coefficient achieved by the corresponding framework with the fixed kernel gradually increases, and the

inspection time increases. This is because the small circular kernel has a small receptive field, which causes poor inspection for large voids. On the contrary, the large circular kernel has a large receptive field, which results in poor inspection for small voids. However, the voids in Region 1 are quite different in size for each chip resistor solder joint. Thus, the fixed kernel size cannot accurately capture the characteristics of the voids with various sizes for various chip resistor solder joints. Compared with the kernel with a fixed size, the proposed ACK greatly improves the inspection ability of the proposed framework in terms of average Dice coefficient, since it is adaptively determined by the area of the largest void in Region 1 for each solder joint, as formulated in (2). Nevertheless, the proposed ACK is a single-scale circular kernel for each solder joint, which cannot well inspect all the voids with quite different sizes in Region 1 of each solder joint. The framework with the proposed MC-ACK achieves the best average Dice coefficient among all the frameworks, since the multiscale and adaptive characteristics of the MC-ACK can enable our proposed framework to focus on the voids with different sizes for different solder joints.

Since the sizes of circular kernels are adaptively determined by (2) for different solder joints, the framework with the proposed ACK costs more inspection time than those with the fixed kernels. Furthermore, the multiscale strategy in the proposed MC-ACK consumes a little more inspection time than the single-scale operation in the proposed ACK. Compared with the acceptable increase of the inspection time, the large improvement of the inspection performance is more concerned by our cooperative company. The fact indicates that the proposed MC-ACK is beneficial for the inspection performance of the proposed framework at a reasonable inspection time.

B. Influences of Gray Adaptive Strategy and SP

The size of the sliding window M contributes to the LPF functions in (8), which influences the evolution of φ . Also, the SP constraint term is a significant term of the energy function in (6). The two issues influence the inspection performance in terms of time efficiency and segmentation accuracy. Here, we conducted an experiment to validate the proposed ACM with the IGA strategy (IGA-ACM) and the proposed ACM incorporating the SP and the IGA strategy (SP-IGA-ACM) by comparing them to the ACM with a fixed sliding window. It is noted that only Region 2 of each solder joint image is utilized for inspection and the MC-ACK scheme is not involved in this experiment.

As indicated in Table II, with the increase of the size of the sliding window, the inspection performance achieved by the proposed framework with the fixed sliding window first increases and reaches the best average Dice coefficient when 9×9 window is utilized for the LPF. Then, the inspection performance decreases slightly when the size of the sliding window is 11×11 . This is because the small window makes the evolution curve of the LPF not sensitive to the voids with low contrast, while the large window makes some redundant information involved in the evolution curve of the LPF.

TABLE I
INSPECTION PERFORMANCE FOR REGION 1 ACHIEVED BY THE PROPOSED FRAMEWORKS WITH DIFFERENT CIRCULAR KERNELS

Size of kernels	9×9	29×29	49×49	99×99	ACK	MC-ACK
Dice	0.4413	0.7287	0.7321	0.7739	0.8453	0.8710
Time(s)	0.204	0.218	0.224	0.229	0.314	0.357

TABLE II
INSPECTION PERFORMANCE FOR REGION 2 ACHIEVED BY THE PROPOSED FRAMEWORKS WITH DIFFERENT SLIDING WINDOWS

Size of sliding windows	3×3	5×5	7×7	9×9	11×11	IGA-ACM	SP-IGA-ACM
Dice	0.8562	0.8872	0.8880	0.8910	0.8907	0.8896	0.9088
Time(s)	2.058	3.195	5.016	7.487	11.208	3.475	3.761

TABLE III
INSPECTION PERFORMANCE ACHIEVED BY THE FRAMEWORKS WITH DIFFERENT COARSE SCHEMES FOR DIFFERENT REGIONS

Methods	Region 1	Region 2
MC-ACK	0.8701	0.5873
SP-IGA-ACM	0.7420	0.9088

TABLE IV
INSPECTION PERFORMANCE ACHIEVED BY THE PROPOSED FRAMEWORK UNDER DIFFERENT QC STANDARDS

QC standard	Ground-truths		Inspection	
	Qualified samples	Unqualified samples	Acc(%)	F1
5%	57	77	94.78	0.9474
10%	106	28	98.51	0.9907
15%	124	10	99.25	0.9960
20%	134	0	100	1

Compared with the slight increase of the inspection performance, the rapid increase of the inspection time is not acceptable.

When the proposed IGA is substituted for the fixed sliding window, the proposed framework with the proposed IGA (IGA-ACM) achieves a little worse inspection performance but significantly less inspection time than the framework with the fixed sliding window of 9×9 . This is because the curve can be rapidly evolved to the optimum since the size of the sliding window is adaptively determined according to the pixel grayscale. Due to the incorporation of the SP strategy, the framework with the SP strategy and the proposed IGA strategy (SP-IGA-ACM) achieves better inspection performance than that with the IGA-ACM and that with the fixed sliding window of 9×9 . Although the SP-IGA-ACM costs a little more inspection time than the IGA-ACM, its inspection speed of 3.761 s per region is acceptable for the real production in our cooperative company, which mostly focuses on the inspection accuracy.

C. Influence of Adaptive Partition

Here, an ablation experiment is designed to discuss the influence of the adaptive partition strategy on the proposed framework, in which the proposed framework with the MC-ACK and that with the SP-IGA-ACM are separately utilized to inspect the voids in Regions 1 and 2.

As indicated in Table III, the framework with the MC-ACK achieves better void inspection performance for Region 1 than that with the SP-IGA-ACM, with the average Dice coefficients 0.8701 versus 0.7420. This is possibly contributed to the reason that the level set function neglects small voids or does not evolve to the edges of the zones with dense voids during evolution. The voids in Region 2 have a much lower contrast to the background compared with those in Region 1, which

will be easily misinspected as the background if the circular convolution operation is utilized for Region 2. The fact can be indicated in the results illustrated in the second volume of Table III. That is, the voids in Region 2 can be better inspected by the framework with the SP-IGA-ACM than by that with the MC-ACK, that is, 0.9088 versus 0.5873 average Dice coefficients. Similar conclusions can be drawn from the results illustrated in the row dimension of Table III.

D. Influence of the QC Standard

In real industries, whether the chip resistor solder joint is qualified or not is determined by the void content. Different QC standards in different companies will influence the two metrics including Acc and F1 score, which are usually employed for evaluation based on classification results. In this section, we conducted an experiment to discuss the influence of QC standard on our proposed framework. Four QC standards, such as 5%, 10%, 15%, and 20%, are utilized for evaluations, which indicate different void contents.

As indicated in Table IV, the numbers of qualified/unqualified chip resistor solder joints are determined as ground truths by an experienced QC worker according to the QC standards. The severer the QC standard is, the relatively worse the inspection performance is achieved by the proposed framework. Specially, the inspection performance under the QC standard of 5% void content is much worse than others. In real industries, the voids inevitably emerge inside the chip resistor solder joint due to the reflow soldering process. Compared with the other QC standards, more chip resistor solder joints have around 5% void content, which indicates that

TABLE V

STATISTICAL RESULTS OF VOID INSPECTION VIA DIFFERENT METHODS

Methods	Acc(%)	F1	Dice	Time(s)
DRLSE	79.10	0.8833	0.5630	14.801
RSF+LoG	79.10	0.8833	0.5965	14.931
GLFIF	79.85	0.8870	0.3429	11.152
LoG	92.54	0.9545	0.8576	0.016
LPF	85.82	0.9177	0.8637	9.628
U-Net	88.06	0.9298	0.7650	0.581
U-Net++	95.45	0.9615	0.8725	0.603
Swin-Unet	88.64	0.9123	0.7801	0.817
Our	98.51	0.9907	0.9069	4.539

slight pixel-level misinspection will result in misclassification. When the QC standard is set to 20% void content, there are no ground-truth unqualified samples, which is meaningless for our proposed framework. In this article, 10% void content is employed as the best QC standard ensured by our cooperative company, which is also the severest standard in the electronic industry.

E. Comparisons of the Existing Inspection Methods

To validate the proposed framework, some existing segmentation methods are employed for comparisons, which are ACMs, such as distance regularized level set evolution model (DRLSE) [29], region-scalable fitting and optimized Laplacian of Gaussian energy level set model (RSF + LoG) [32], LPF [27], and global and local fuzzy image fitting model (GLFIF) [33], the traditional method, such as LoG [12], and deep learning, such as U-Net [19], UNet++ [34], and Swin-Unet [35]. For fair comparisons, the values of parameters with the same influence are set as the same for all the ACMs, such as $v = 0.02 \times 255^2$, $\mu = 2$, $\Delta t = 0.02$, and the convergence condition is $|(S_i - S_{i-2}/S)| < 10^{-5}$. For some unique parameters, we refer to the parameters recommended in the corresponding published articles. It is noted that all the methods except the three deep learning methods incorporate the void fine inspection proposed in Section III-D.

Two qualified chip resistor solder joints and two unqualified ones are demonstrated in Fig. 5 for visual evaluation of different inspection methods, in which original images, ground truths, and void inspection results via different methods are involved. Also, the void content for each image is illustrated below the image. As illustrated in Fig. 5, many voids, especially obscure voids, are not inspected by the DRLSE, RSF + LoG, and GLFIF, which results that these methods tend to distinguish unqualified samples as qualified. The LoG, LPF, and U-Net can inspect most of the voids, which are superior to the DRLSE, RSF + LoG, and GLFIF. However, some error inspections emerge in the results achieved by the three deep learning methods, since they do not incorporate void fine inspection. The inspection results achieved by the proposed framework are highly close to the ground truths in terms of visual effect and void content.

To completely evaluate the proposed framework, the statistical analysis of void inspection for 134 X-rayed images of chip resistor solder joints is performed for different methods, as shown in Table V. Here, 10% void content is

employed as the best QC standard for evaluating the inspection accuracy, which is ensured by our cooperative company.

The DRLSE uses local edge information to attract the level set function to the boundaries of the targets, which is sensitive to noise and initial contours. The local edge information of the targets with weak edges is not salient, which will result that the level set function possibly ignores these targets during evolution. Since most of the voids in the chip resistor solder joint images are obscure, the DRLSE achieves a poor pixel-level inspection performance of 0.5630 average Dice coefficient.

The RSF + LoG, a region-based ACM, introduces the optimized Gaussian Laplacian energy term into the energy function to improve the robustness of the model to noise and initial contour. Thus, it is better than the GLFIF and DRLSE in terms of average Dice coefficient. However, since the energy term based on local binary fitting is only suitable for classifying an image into two categories, the RSF + LoG is not suitable for multitarget and multiscale image segmentation tasks [36]. The fact results in its fairly poor inspection for the voids in the chip resistor solder joints.

The GLFIF achieves the worst average Dice coefficient of 0.3429 among all the methods. This is because it is an ACM that combines a global term via binary fitting with a fitting term composed of global fuzzy fitted and local fuzzy fitted image. Relatively, the global term performs a larger contribution to the energy function than the fitting term. Thus, the GLFIF is not suitable for multitarget voids with different contrasts.

Gaussian filter and second derivative involved in the LoG are beneficial for the segmentation of the targets with weak edges, which also have a fair anti-noise ability. So, the LoG achieves a fairly good performance in terms of 92.54% accuracy and 0.9545 F1 score. Also, its inspection speed can reach 0.016 s per image. The two facts make it widely applied to void inspection for BGA components. However, the LoG is not sensitive to the obscure voids, which results that some obscure voids cannot be correctly inspected. Thus, it achieves a low average Dice coefficient of 0.8576, which indicates that it is poor in the pixel-level inspection. As illustrated in Fig. 5, there are some differences between the inspected results achieved by the LoG and the ground truths. Especially, some large voids are missed for inspection, which will influence the component reliability and even result in the component failure. So, this potential failure risk is not acceptable in real industries, although its inspection accuracy seems fairly good.

The LPF employs an LPF energy function for curve evolution, in which two prefitting functions are defined by the average intensities in a local window. Compared with the ACMs with binary fitting and edge-based ACMs, the LPF can make full advantages of the neighbor information of the edges due to the two fitting functions. So, it achieves a fairly good average Dice coefficient of 0.8637, which indicates its fairly good pixel-level inspection ability. However, due to the fixed size of its local window, it does not perform a good inspection for the chip resistor solder joint image with salient background difference.

Among the three deep learning methods, the U-Net achieves the worst performance with an average Dice coefficient of

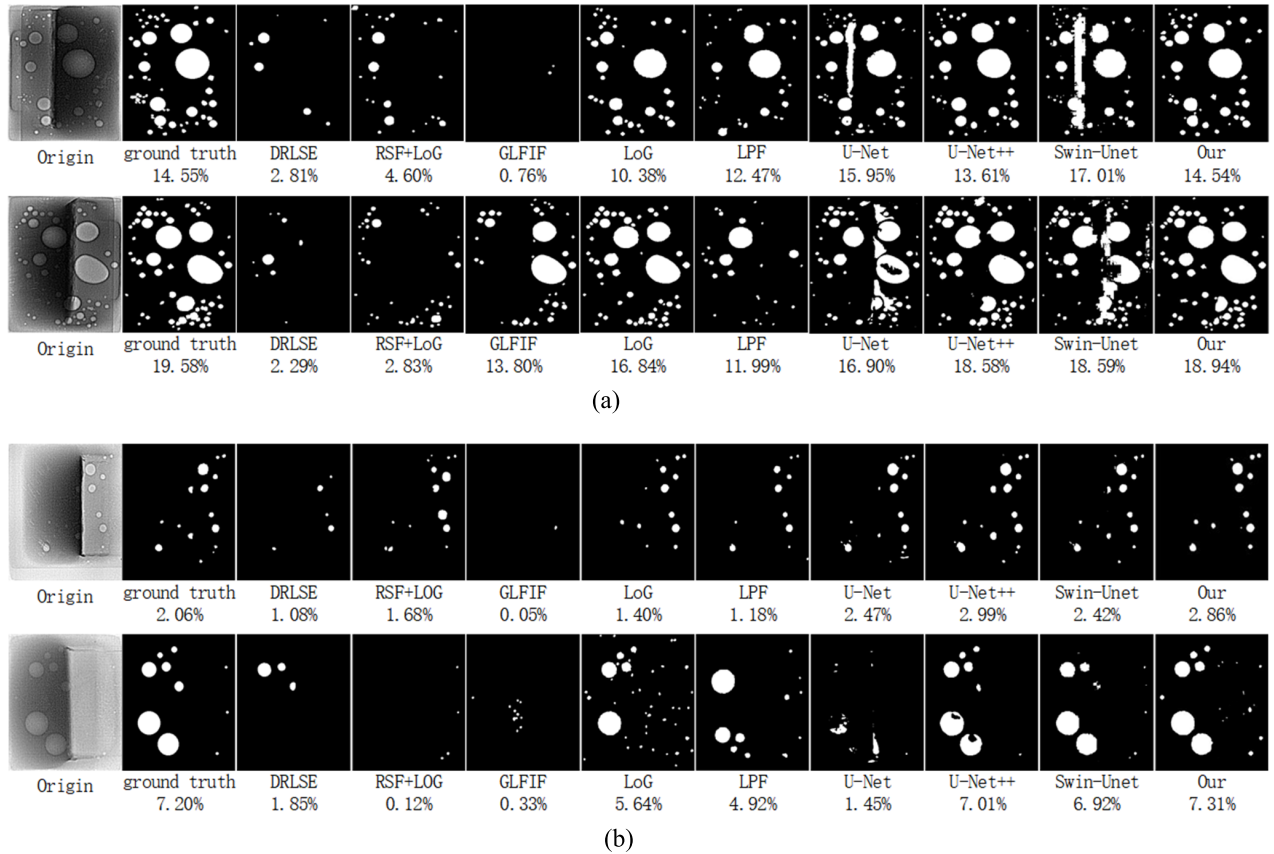


Fig. 5. Visual results of the void inspection via different methods. (a) Unqualified samples. (b) Qualified samples.

0.7650, which is also worse than the LPF. This is because its optimal depth is a priori unknown and its skip connections impose an unnecessarily restrictive fusion scheme [34]. Since the UNet++ introduces a built-in ensemble of U-Nets of varying depths and redesigns skip connections by densely connecting the decoders of the constituents U-Nets at the same resolution, it can improve segmentation performance for varying size objects and flexible feature fusion in decoders [34]. So, it can achieve the best inspection performance with an average Dice coefficient of 0.8725 among all the methods except the proposed method. The Swin-Unet introduces the state-of-the-art transformer scheme to design a transformer-based U-Net network structure for segmentation, which can well learn global and long-range semantic information interaction [35]. So, it has an improvement of inspection performance over the U-Net. However, the Swin-Unet meanwhile requires a huge number of data and time for pretraining, which results that it achieves worse inspection performance than the UNet++. In general, as data-driven methods, the deep learning methods achieve worse inspection performance compared with the proposed framework due to the limitation of the number of chip resistor solder joints provided by our cooperative company. Although they spend a little time inspecting a chip resistor solder joint, the networks should be well-trained before their applications, which will cost much time.

Our proposed framework achieves the best performance in terms of 0.9069 average Dice coefficient, 0.9907 F1 score,

and 98.51% accuracy among all the methods. Also, as illustrated in Fig. 5, all the voids are almost inspected by the proposed framework, which demonstrates the excellent pixel-level inspection ability of the proposed framework. The facts are mainly contributed to three proposed schemes, which are adaptive partition, MC-ACK, and SP-IGA-ACM. The adaptive partition can adaptively divide the chip resistor solder joint into two regions, which can be separately dealt with by the appropriate inspection schemes. The multiscale and adaptive characteristics of the MC-ACK can make the inspection focus on the dense voids with different sizes in Region 1. The SP constraint term in the SP-IGA-ACM is beneficial for the inspection of overlapping voids in Region 2.

Compared with the other ACMs, our proposed framework costs about a half and even a third of the inspection time to inspect a chip resistor solder joint. This is contributed to the IGA strategy involved in the proposed framework, which utilizes a sliding window with an adaptive size to achieve the LPF energy term. However, compared with the LoG, the inspection speed of 4.539 s per image for the proposed framework is relatively lower. This speed limitation is contributed to the iterative mechanism of curve evolution in the ACM-based methods. Although the inspection speed of 4.539 s per image is time-efficiently acceptable for our cooperative company, we can further accelerate the inspection of the proposed framework utilizing two strategies. One is to apply parallel processing to the proposed framework. We have attempted to employ the parallel processing library

(multiprocessing) in Python to perform parallel processing for our framework on our experimental platform of a computer with Windows10 64-bit, six-core Intel i5-10600KF CPU at 4.10 GHz. Parallel experiments show that our framework can accelerate the inspection speed up to 0.870 s per image, which is nearly 6 times of the former inspection speed without parallel processing. As far as we know, the number of cores of the CPU greatly helps to improve the parallel efficiency [37]. Therefore, if the funds are enough, the processing time can be further reduced with more advanced equipment. Second, the Python has low efficiency in execution compared to C-like programming language in many computational tasks [38]. So, we can substitute C-like language with multiple threads for the Python in the future work, which will significantly improve the execution efficiency of our proposed method. In general, it is expected that the processing time of the proposed method will be possibly reduced to less than 0.2 s when all the above acceleration schemes are employed in the future.

V. CONCLUSION

The void content in the chip resistor solder joint influences the service life and reliability of the electronic device, which is necessarily evaluated in real SMT industries. In this article, an adaptive hybrid framework is proposed for multiscale void inspection of chip resistor solder joints. To automatically achieve good inspection performance at a reasonable speed, several novel adaptive schemes are incorporated into the proposed framework, which are adaptive partition, MC-ACK, and SP-IGA-ACM for void coarse inspection, and the strategies of shape factor and average gray for void fine inspection.

Experimental results indicate that the multiscale and adaptive characteristics of the MC-ACK contribute to the proposed framework for inspecting the voids with different sizes in the region of solder beneath the chip resistor body. The SP constraint term and gray adaptive strategy incorporated into the framework can well inspect the voids in the region of the solder fillet at a reasonable inspection speed. Comparison experiments indicate that the proposed framework achieves better inspection performance than some existing LoG-based, ACM-based, and deep-learning-based methods, in terms of 0.9069 average Dice coefficient, 0.9907 F1 score, and 98.51% accuracy.

One of the limitations of our proposed framework is a relatively low inspection speed of 4.539 s per image. We can accelerate the proposed framework through implementing it by parallel processing and C-like programming in the future. Another limitation is the Otsu method, a global thresholding method, utilized for inspecting the voids inside the solder beneath the chip resistor body, which does not perform an excellent segmentation task for the filtered image via the guided filtering. In the future, the local thresholding strategy will be studied to further improve the inspection performance of the proposed framework. Moreover, it is not an optimal strategy to simply adopt a mean value of kernels for large-scale and small-scale convolutions to achieve the circular kernel for the midscale convolution. We can design a better strategy to achieve a more accurate circular kernel of the midscale

convolution adaptively and high-efficiently according to many statistical results. This will be performed after a large number of chip resistor solder joint samples are collected over a long period.

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